

3.5.2 Nuclear Energy

- **Mass and energy**

Appreciation that $E = mc^2$ applies to all energy changes.

Simple calculations on mass difference and binding energy.

Atomic mass unit, u; conversion of units;

$$1 \text{ u} = 931.3 \text{ MeV}$$

Graph of average binding energy per nucleon against nucleon number.

Fission and fusion processes.

Simple calculations from nuclear masses of energy released in fission and fusion reactions.

- **Induced fission**

Induced fission by thermal neutrons; possibility of a chain reaction; critical mass.

The functions of the moderator, the control rods and the coolant in a thermal nuclear reactor; factors affecting the choice of materials for the moderator, the control rods and the coolant and examples of materials used; details of particular reactors are not required.

- **Safety aspects**

Fuel used, shielding, emergency shut-down.

Production, handling and storage of radioactive waste materials.